

μOS++ III SMP

Raspberry Pi Zero 2 W hardware tests, step by step

Raspberry Pi Zero 2 W — running one test on the board, step by step

This page runs one μ OS++ test on a Raspberry Pi Zero 2 W through OpenOCD, by hand, for the 32-bit (AArch32) and the 64-bit (AArch64) port. It does the same thing the test scripts (`hw.sh` , `run-hw.sh`) do, written out as plain commands.

The probe is a SEGGER J-Link, used through OpenOCD. (An Olimex ARM-USB-OCD also works; see the end of the page.)

What happens, in short

1. The board is reset through its watchdog, and boots again from its SD card. Core 0 runs the SD card's kernel; cores 1–3 wait in the firmware's own loop.
2. OpenOCD stops the four CPU cores.
3. OpenOCD turns semihosting on, so the test can print through the probe.
4. OpenOCD copies the test program into the board's RAM.
5. OpenOCD starts **core 0** at the program's first instruction, and lets cores 1–3 go on where they were: still waiting in the firmware's loop.
6. When the test's kernel is ready, it releases cores 1–3 itself.
7. The test runs and prints `RESULT: PASS` (or `FAIL`).

Each test starts with the watchdog reset of step 4, so the tests can be run one after another. (On 2026-10-09 all 14 tests but `usb_test` passed this way, back to back, on both ports.) If a test hangs, power-cycle the board.

The SD card

Cores 1–3 must still be in the firmware's loop when step 6 runs, so the SD card must boot a kernel that does **not** start them, and it must boot the same width as the port, because the two ports wake the cores differently:

Port	SD card <code>config.txt</code>	The kernel wakes cores 1–3 by writing	Cores 1–3 wait at (after step 4)
AArch32	<code>arm_64bit=0</code>	their mailbox 3	<code>pc=0x7a</code> , Thumb, Hypervisor
AArch64	<code>arm_64bit=1</code> (e.g. <code>\$A64/test/boards/rpi-zero-2w/hw-park/</code>)	their release word at <code>0xd8 + 8*core</code>	<code>pc=0x7c</code> , EL2H

So switch the SD card (or its `config.txt` and kernel) when you switch ports. To check it, after a step-4 reset and a 12 s wait:

```
$OPENOCD -s $CFG -f $CFG/openocd-jlink-rpi3.cfg -c "init" \  
-c "targets bcm2837.cpu1" -c "halt" -c "reg pc" -c "resume" \  
-c "shutdown"
```

`pc` must be the value in the table (the same for `cpu2`, `cpu3`). If a core is anywhere else, the SD card's kernel has started it, and these steps will not work: use another card, or the previous flow, kept in each port's `test/boards/rpi-zero-2w/hw.sh.bak` (it starts all four cores and first clears the `__smp_spin` table in RAM).

Step 0 — set the paths

```
WORK=/home/dan/Desktop/tmp_tests          # the folder that holds the repositories  
K=$WORK/micro-os-plus-iii                 # kernel  
A32=$WORK/micro-os-plus-iii-aarch32      # 32-bit port  
A64=$WORK/micro-os-plus-iii-aarch64      # 64-bit port  
BUILD=$K/tests/build  
OPENOCD=$HOME/.local/xPacks/@xpack-dev-tools/openocd/0.12.0-7.1/.content/bin/openocd
```

Step 1 — build the tests

```
cd $K/tests  
xpm install --config aarch32-rpi-zero-2w-cmake-gcc-debug  
xpm run prepare --config aarch32-rpi-zero-2w-cmake-gcc-debug  
xpm run build --config aarch32-rpi-zero-2w-cmake-gcc-debug
```

For the 64-bit port, write `aarch64` instead of `aarch32`.

The programs for the board are the files ending in `-hwd`, in `$BUILD/aarch32-rpi-zero-2w-cmake-gcc-debug/platform-bin/port-tests/test/` (or `aarch64-...`). The tests are: `smp_test0` ... `smp_test4`, `smp-mat-test`, `smp-mat-sdcard-test`, `smp-num-test`, `smp-pipeline-test`, `smp-pro-cons-test`, `sd_test`, `usb_test`, `cmsis-os-validator`, `mutex-stress`, `rtos-apis`.

AArch32 (32-bit port)

Step 2 — choose the test

```
D=$BUILD/aarch32-rpi-zero-2w-cmake-gcc-debug/platform-bin/port-tests/test
CFG=$A32/test/boards/rpi-zero-2w
ELF=$D/smp_test0-hwd
```

Nothing has to be read from the ELF: the program always starts at `0x1003c`.

Step 3 — open the serial console (optional)

In another terminal:

```
tio -b 115200 /dev/ttyACM0
```

OpenOCD never uses this port, so it can stay open all the time.

Step 4 — reset the board

```
$OPENOCD -s $CFG -f $CFG/openocd-jlink-rpi3.cfg -c "init" \
-c "targets bcm2837.cpu0" \
-c "halt" \
-c "mww 0x3f100024 0x5a000001" \
-c "mww 0x3f10001c 0x5a000020" \
-c "shutdown"
```

- `-s $CFG -f $CFG/openocd-jlink-rpi3.cfg` — the J-Link and the board description (4 Cortex-A53 cores, JTAG at 1000 kHz). The file only describes the hardware; it does not connect by itself.
- `-c "init"` — connect to the board through the probe. Every OpenOCD command below needs it first.
- `targets bcm2837.cpu0` — talk to core 0.
- `halt` — stop core 0.
- `mww 0x3f100024 0x5a000001` — set the watchdog timer to (almost) zero.
- `mww 0x3f10001c 0x5a000020` — tell the watchdog to reset the chip when it expires. The board reboots.
- `shutdown` — OpenOCD exits.

The `Error: Invalid ACK (0) in DAP response` lines at the end are expected: the chip resets while the probe is still talking to it.

Step 5 — wait for the board to boot

```
sleep 12
```

Step 6 — load the test and start it

This is one OpenOCD command. The options run in order, from top to bottom.

```
$OPENOCD -s $CFG -f $CFG/openocd-jlink-rpi3.cfg -c "init" \  
-c "targets bcm2837.cpu0" -c "halt" \  
-c "targets bcm2837.cpu1" -c "halt" \  
-c "targets bcm2837.cpu2" -c "halt" \  
-c "targets bcm2837.cpu3" -c "halt" \  
-c "targets bcm2837.cpu0" -c "arm semihosting enable" \  
-c "targets bcm2837.cpu1" -c "arm semihosting enable" \  
-c "targets bcm2837.cpu2" -c "arm semihosting enable" \  
-c "targets bcm2837.cpu3" -c "arm semihosting enable" \  
-c "targets bcm2837.cpu0" -c "load_image $ELF" \  
-c "targets bcm2837.cpu0" -c "reg cpsr 0x600001da" -c "resume 0x1003c" \  
-c "targets bcm2837.cpu1" -c "resume" \  
-c "targets bcm2837.cpu2" -c "resume" \  
-c "targets bcm2837.cpu3" -c "resume"
```

What each part does:

1. **Stop the 4 cores** (`targets` ... `halt` , for cores 0–3), so nothing runs while the program is copied.
2. **Turn semihosting on for each core.** The test prints by stopping the core for a moment; OpenOCD sees the stop, prints the text and lets the core go on. Every core needs it, because any core can print once it runs.
3. **Copy the program into RAM** (`load_image $ELF` , through core 0).
4. **Start core 0** at the program's entry:
5. `reg cpsr 0x600001da` — put the core in 32-bit mode: the low bits `0x1a` select HYP (hypervisor) mode, and `0x1c0` masks the aborts, IRQs and FIQs;
6. `resume 0x1003c` — run from the entry point.
7. **Let cores 1–3 go on where they were** (`resume` with no address): back in the firmware's loop, waiting on their mailbox 3.

Core 0 starts the system. When the kernel is ready, the test wakes each of cores 1–3 (`release_one()` in `test/boards/rpi-zero-2w/src/smp.cpp`): it first writes the core's start address into its `__smp_spin` slot, then writes `__start` into the core's mailbox 3. The core leaves the firmware's loop, runs `__start` , and finds its `__smp_spin` slot already written — so whatever the previous program left in RAM there does not matter, and OpenOCD does not have to clear it.

Step 7 — watch the result

The test's messages appear in this OpenOCD window. It passed when it prints `RESULT: PASS`. Then press **Ctrl-C** to stop OpenOCD. An SMP test shows that cores 1–3 started, e.g. `join: c1=3 c2=3 c3=3`.

How long to wait at most: `smp_test0` – `smp_test3` 120 s; `smp_test4`, `usb_test`, `cmsis-os-validator`, `mutex-stress`, `rtos-apis` 300 s; `sd_test` 450 s; `smp-num-test`, `smp-pipeline-test`, `smp-pro-cons-test` 600 s; `smp-mat-test`, `smp-mat-sdcard-test` 900 s. Printing is slow over JTAG, so the chatty tests take long.

Step 8 — the next test

Go back to step 2: the reset of step 4 puts cores 1–3 back in the firmware's loop. If a test hung, power-cycle the board first.

AArch64 (64-bit port)

The same steps; only these things change:

	AArch32	AArch64
SD card	<code>arm_64bit=0</code>	<code>arm_64bit=1</code>
Config folder <code>CFG</code>	<code>\$A32/test/boards/rpi-zero-2w</code>	<code>\$A64/test/boards/rpi-zero-2w</code>
Entry point	<code>0x1003c</code>	<code>0x80000</code>
Start core 0	<code>reg cpsr 0x600001da</code> , <code>resume</code> <code>0x1003c</code>	<code>reg pc 0x80000</code> , <code>resume</code>
Cores 1–3	<code>resume</code> where they are	<code>resume</code> where they are
JTAG speed	1000 kHz	4000 kHz

Step 2 (AArch64)

```
D=$BUILD/aarch64-rpi-zero-2w-cmake-gcc-debug/platform-bin/port-tests/test
CFG=$A64/test/boards/rpi-zero-2w
ELF=$D/smp_test0-hwd
```

Nothing to read from the ELF: the entry is always `0x80000`.

Step 4 (AArch64) — reset the board

```
$OPENOCD -s $CFG -f $CFG/openocd-jlink-rpi3.cfg -c "init" \  
-c "targets bcm2837.cpu0" \  
-c "halt" \  
-c "mww 0x3f100024 0x5a000001" \  
-c "mww 0x3f10001c 0x5a000020" \  
-c "shutdown"
```

The same as for AArch32, with the 64-bit port's config folder.

Step 5 (AArch64)

```
sleep 12
```

Step 6 (AArch64) — load the test and start it

```
$OPENOCD -s $CFG -f $CFG/openocd-jlink-rpi3.cfg -c "init" \  
-c "targets bcm2837.cpu0" -c "halt" \  
-c "targets bcm2837.cpu1" -c "halt" \  
-c "targets bcm2837.cpu2" -c "halt" \  
-c "targets bcm2837.cpu3" -c "halt" \  
-c "targets bcm2837.cpu0" -c "arm semihosting enable" \  
-c "targets bcm2837.cpu1" -c "arm semihosting enable" \  
-c "targets bcm2837.cpu2" -c "arm semihosting enable" \  
-c "targets bcm2837.cpu3" -c "arm semihosting enable" \  
-c "targets bcm2837.cpu0" -c "load_image $ELF" \  
-c "targets bcm2837.cpu0" -c "reg pc 0x80000" -c "reg pc" -c "resume" \  
-c "targets bcm2837.cpu1" -c "resume" \  
-c "targets bcm2837.cpu2" -c "resume" \  
-c "targets bcm2837.cpu3" -c "resume"
```

The same parts as for AArch32, except:

- **Start core 0:** `reg pc 0x80000` sets its program counter to the entry, `reg pc` reads it back (OpenOCD prints `pc (/64): 0x0000000000008000`, a check only), `resume` runs it. A 64-bit core keeps the mode it was stopped in (EL2), and the startup code reads that mode, so there is no CPSR to set.
- **Cores 1–3** wait in the firmware's loop on their release word (`0xd8 + 8*core`), not on a mailbox; the test writes `_start` there after it has written their `_smp_spin` slot.

Steps 3, 7 and 8 are the same as for AArch32.

With the Olimex ARM-USB-OCD instead of the J-Link

In steps 4 and 6, write `openocd-olimex.cfg` instead of `openocd-jlink-rpi3.cfg`. Nothing else changes.

If something goes wrong

- `LIBUSB_ERROR_BUSY` — another OpenOCD still holds the probe: `pkill -9 -f openocd`.
- `Invalid ACK (0) in DAP response` in step 6, then the cores cannot be examined — the JTAG link dropped, not the test. Lower the speed: add `-c "adapter speed 1000"` right after the config file, and check the board's power supply.
- **The cores halt in the wrong state** (AArch32 card on an AArch64 test, or the other way round: `ARM state ... Hypervisor` instead of `AArch64 ... EL2H`) — the SD card boots the other width, and the test times out; see "The SD card".
- **Nothing prints** — power-cycle the board and start again from step 4.